Raunit Kohli

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WORK EXPERIENCE

Rocket Software **Boston, MA**

Machine Learning Software Engineer III 09/2024 – Present

- Architect end-to-end Identity and Access Management (IAM) for Rocket EVA™ — ship ground-up OIDC-based authentication, cross-language runtime executable for real-time customer license validation, MCP management layer authorization and registration UI, and cross-team integration with mainframe-scoped RBAC model in one quarter.
- Author customer-facing technical documentation and train sales engineers to independently configure and demo the platform for prospective clients, cutting non-technical installation-and-demo time from 3 days to 2 hours.
- Deploy secure live customer cloud environments on AWS (EC2, Cognito) using Terraform, analyze error logs to resolve user and QA reported product-behavior questions, and ship features and bug fixes directly against pilot feedback.
- Create and deliver AI foundations workshops for company apprenticeship program at global offices, training 95+ graduates to apply techniques daily; lead coordination of enterprise-wide AI enablement program for 1800+ engineers.
- Build SAS-to-Python code translation tool on specialized team using DSPy ReAct agent, collaborating on Agentic Context Engineering (ACE) optimization framework; chains 3 per-input agentic workflows for architecture design, test generation, and MLflow tracked loop that runs tests and repairs attempted generation against reported failures.
- Design and implement retrieval-based grounding and citation generation for enterprise RAG assistant, using Gaussian Mixture Model to dynamically rerank cross-encoder and BM25 similarity scores, yielding 80% more relevant references.
- Develop few-shot prompt-engineering framework to translate natural language queries into executable SQL, integrating multi-step sub-agent workflows and regex-based syntax validation, enhancing table-based document retrieval and QA.

Robotics Using Bayesian Inference Lab – UC San Diego **San Diego, CA**

Machine Learning Research Consultant (Published) 09/2024 – Present

- Architect composable data pipeline for cleaning kinematic agent-interaction data and training models to identify proxemic predictors of hand-labeled pro-social behavior; present explore-exploit dynamics findings at IEEE ICDL 2025.
- Shape reward functions for deep RL policy-optimization algorithms using importance-ranked homebase-derived proxemic features, modeling bio-influenced exploration-regulation dynamics for robotic behavior in MuJoCo simulation.
- Founding engineer hired for multi-million-dollar AI-powered classroom project; built PTPv2-synchronized multi-modal embedded system for secure data streaming and recording with custom Arduino firmware for OpenBCI biophysical sensors and high-precision Basler camera integration.

Staff Machine Learning Research Engineer (Published) 01/2021 – 09/2024

- Spearheaded implementation of reinforcement learning algorithms in PyTorch for robotic PiRat over ROS2 with sim-to-real transfer of autonomous behavioral routines to physical robot velocity commands, utilizing extensive neural and positional data from 2500+ rat experiments to mimic and predict multi-agent interactions.
- Developed live multi-agent pose estimation system using SLEAP neural networks with multi-threaded architecture.
- Enhanced and scaled production-deployed computer vision-based live pose-estimation Software in C++ to validate statistical accuracy of Kinect Tracker, introducing new features and parameter logging, yielding 3x processing speed.

HIGHLIGHTED PROJECTS

VLM-Guided Hierarchical RL in robot simulation (WPI, 2025) – Hierarchical PPO/SAC for Fetch robot in Meta Research's Habitat-Lab with VLM high-level planner for autonomous navigation and manipulation from language instructions.

Smart Scrum Master – Rocket.Build 2025 "Most Innovative" Winner – LLM-powered Agile assistant and MCP Server optimizing Jira workflows via reasoning agent, winning 1 of 6 awards from 650+ global company projects.

Aspera Workflows Automated Test Suite (IBM Engineering Intern, 2023) – Built complete 780+ test automated regression framework in Ruby for IBM Aspera on Cloud API, identifying 15+ critical production bugs across all endpoints.

TECHNICAL SKILLS

Languages: Python, C++, Ruby, Java, SQL, Bash; **Libraries:** PyTorch, DSPy, MLflow, HuggingFace, OpenCV
Domains: Policy-Optimization Algorithms, Multi-Agent Systems, Agentic AI, Natural Language Processing, Computer Vision, Markov Decision Processes, Signal Processing, Pose Estimation, Data Analysis (Pandas/NumPy/Scikit-learn)
Tools: AWS, Docker, Kubernetes, Postgres, Kafka, ROS2, MuJoCo, Git/GitHub, Atlassian (Jira, Confluence, Bitbucket)

EDUCATION

University of California – San Diego **09/2020 – 12/2023**

Bachelor of Science, Machine Learning and Neural Computation Major GPA: 3.91/4.0

Minor: Computer Science; **Program of Concentration:** Philosophy

Awards: President & Recruitment Advisor – Phi Sigma Pi National Honor Fraternity; Provost Honors (Warren College)